

FLIGHT CONTROLLER

ARKV6X

Triple-IMU FMUv6X autopilot module on the Pixhawk Autopilot Bus

EXPOSED INTERFACES

CAN UART I2C SPI PWM USB ETH RC PPM

The USA-built ARKV6X flight controller is based on the FMUv6X and Pixhawk Autopilot Bus (PAB) open source standards. An STM32H743 processor reads three time-synchronised IMUs, each on its own SPI bus and its own switched 3.3 V rail, making data averaging, voting and filtering possible rather than leaving the estimator with a single point of failure.

Conforming to the PAB form factor, it drops into any PAB-compatible carrier — the ARK PAB Carrier, ARK Jetson PAB Carrier or ARK Jetson PAB V3 — so one autopilot module moves between airframes and compute payloads.

01 KEY FEATURES

- Three synchronised IMUs, each independently powered
 - 2 × TDK InvenSense ICM-42688-P (SPI2, SPI3)
 - 1 × TDK InvenSense IIM-42652 industrial IMU (SPI1)
- Bosch BMP390 barometer and BMM150 magnetometer on internal I2C4
- Four independently switched 3.3 V sensor rails — one per sensor domain
- Cypress FM25V02A 256 kbit FRAM for non-volatile parameter storage
- 1 W (25 Ω) sensor heater under PWM closed-loop control
- Supercapacitor-backed real-time clock
- microSD logging on a 4-bit SDMMC interface

02 SPECIFICATIONS

PROCESSOR & MEMORY

MCU	ST STM32H743IIK6, 176-UFBGA
Flash / RAM	2 MB / 1 MB
FRAM	FM25V02A, 256 kbit, 40 MHz SPI
Oscillators	16.000 MHz + 32.768 kHz ±20 ppm

ELECTRICAL

Supply voltage	4.5 – 5.5 V (5 V nominal)
Supply current	500 mA max at 5 V
— main system	300 mA
— sensor heater	200 mA at 5 V (25 Ω, PWM)
Operating temperature	-40 °C to +85 °C

MECHANICAL

Dimensions	36 × 29 × 5 mm
Weight	5.0 g
Mounting	4 × Ø2.3 mm on 31.4 × 24.4 mm



ARKV6X

AT A GLANCE

Standard	FMUv6X / PAB
MCU	STM32H743IIK6
IMUs	3 (2 + 1 industrial)
Baro / Mag	BMP390 / BMM150
Supply	4.5 – 5.5 V, 500 mA
Operating temp	-40 to +85 °C
Interface	100 + 50 pin PAB
Dimensions	36 × 29 × 5 mm
Weight	5.0 g
Hardware rev	Rev 4.1

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- Blue UAS Framework selected
- FMUv6X open standard
- Pixhawk Autopilot Bus DS-010 standard

ORDERING & RESOURCES

STORE	arkelectron.com/product/arkv6x/
DOCS	docs.arkelectron.com/products/flight-controller/arkv6x
CAD	github.com/ARK-Electronics/ark-hardware
BUNDLE	ARKV6X Bundle — with ARK PAB Carrier

03 PIXHAWK AUTOPILOT BUS INTERFACE

PAB CONNECTOR	HIROSE PART	SIGNALS CARRIED
PAB X1 (J2)	DF40C-100DP-0.4V, 100 pos	8 × PWM · CAN1 / CAN2 · 8 × UART (see serial map) · I2C1–I2C3 · USB with VBUS sense · PPM and RC input · SWD · safety switch, buzzer, armed · 3 × power-input sense · ADC · 5 V input
PAB X2 (J3)	DF40C-50DP-0.4V, 50 pos	Ethernet RMII with power enable · external SPI6 (2 × CS, reset, 2 × DRDY) · SPIX_SYNC · 17 reserved pins

04 SENSOR BUS MAP

DEVICE	INTERFACE	ADDRESS / SELECT	FUNCTION
ICM-42688-P (U12)	SPI2	nCS1	6-axis accelerometer / gyroscope
ICM-42688-P (U13)	SPI3	nCS1	6-axis accelerometer / gyroscope
IIM-42652 (U9)	SPI1	nCS1	Industrial 6-axis accel / gyro
BMP390 (U11)	I2C4	0x76	Barometric pressure
BMM150 (U10)	I2C4	0x10	3-axis magnetometer

05 CARRIER BOARD OPTIONS

ARK CARRIER BOARD	AVIONICS I/O	ADDS
ARK Pixhawk Autopilot Bus Carrier	JST-GH	Full Pixhawk-standard breakout, 100 Mbps Ethernet, dual power
ARK Jetson PAB Carrier	JST-GH	NVIDIA Jetson Orin compute on Pixhawk-standard ports
ARK Jetson PAB V3	Pico-Clasp	Jetson Orin compute, triple power inputs, Ethernet switch
ARK VOXL2 RTK PAB Carrier	Molex	ModalAI VOXL2 compute; Blue UAS Framework listed

06 ALTERNATIVES IN THE ARK AUTOPILOT LINE

AUTOPILOT	INERTIAL SENSING	POWER INPUT	WHEN TO CHOOSE IT
ARKV6X Extended Range	3 IMUs, voting	5 V via carrier	Higher angular-rate range (IIM-42653)
ARKV6S	1 industrial IMU	5 V via carrier	Cost-optimised, no IMU redundancy
ARK FPV	1 industrial IMU	3–12S direct	Standalone 30.5 mm build, 12 V payload rail

07 SERIAL REFERENCE & FIRMWARE SUPPORT

Serial Port Mapping

UART	DEVICE	PORT
USART1	/dev/ttyS0	GPS
USART2	/dev/ttyS1	TELEM3
USART3	/dev/ttyS2	Debug Console
UART4	/dev/ttyS3	UART4 & I2C
UART5	/dev/ttyS4	TELEM2
USART6	/dev/ttyS5	PX4IO/RC
UART7	/dev/ttyS6	TELEM1
UART8	/dev/ttyS7	GPS2

Firmware & software support

- PX4 Autopilot — ships by default; ArduPilot supported
- Board-to-board pinout follows the DS-010 Pixhawk Autopilot Bus Standard

Carrier required. All I/O is presented on the two PAB board-to-board connectors; wiring connectors are provided by the carrier board.

ARK-DS-ARKV6X | Issue 2026-08 | Compiled from the ARKV6X Production schematic Rev 4.1 (13 Dec 2023), the ARKV6X Production bill of materials, the ARKV6X production STEP model, and ARK Electronics published documentation. Specifications are subject to change without notice; confirm against current published documentation before design commitment. ARK Electronics products are designed and manufactured in the USA. © 2026 ARK Electronics LLC.